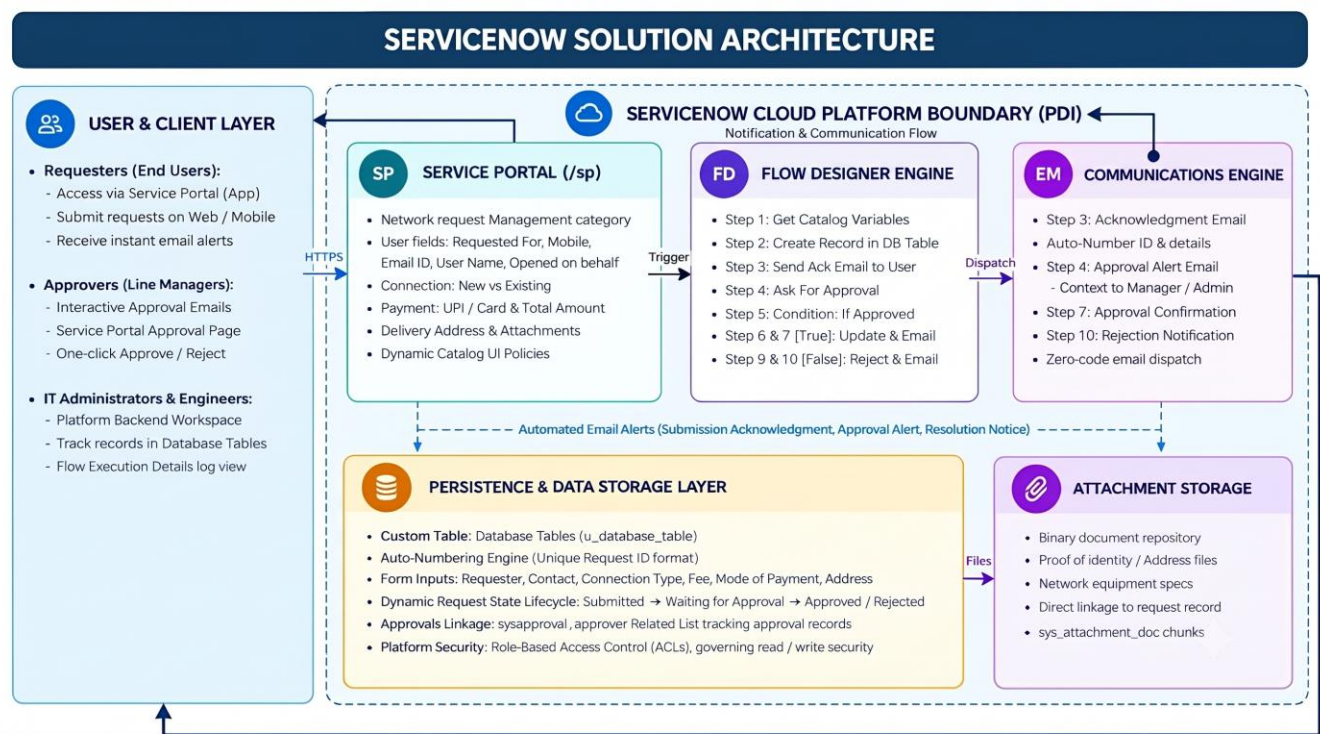


Project Design Phase

Solution Architecture

Date	03 October 2026
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Project Name	Automated Network Request Management in ServiceNow

1. Technical Solution Architecture Diagram



Architecture Overview: The solution architecture defines the multi-tiered structure implemented in the ServiceNow Personal Developer Instance (PDI). It connects the user experience on the Service Portal (/sp) with backend relational data models and zero-code workflow automation. Requests flow from client-side intake forms, through Flow Designer logic for automated validation and manager approval, into the custom Database Tables (u_database_table) schema, with milestone notifications dispatched back to requesters via platform email services.

2. Architectural Tiers & Component Specifications

Layer / Tier	ServiceNow Component	Role & Implementation Details
Presentation / Client Tier	Service Portal (/sp) & Service Catalog	• Hosts 'Network Request' catalog item under 'Network request Management'. • Dynamic Catalog UI Policies show/hide fields conditionally. • Captures: Requested For, Opened on behalf of, User Name, Phone, Email, Mobile, Type of Connection (New/Existing), Total Amount, Mode of Payment (UPI/Card), Address, and Attachments.
Business Logic Tier	Flow Designer (10 Steps)	• Trigger: Service Catalog on form submission. • Step 1: Get Catalog Variables from Network Request. • Step 2: Create Record in Database Tables (u_database_table). • Step 3: Send instant Acknowledgment Email to Requester. • Step 4: Ask For Approval on Database Tables. • Step 5: Evaluates approval condition (true = Approved, false = Rejected). • True Branch (Steps 6-7): Update Record (Approved) + Send Approval Email. • False Branch (Steps 9-10): Update Record (Rejected) + Send Rejection Email.
Persistence Tier	Custom Table: Database Tables (u_database_table)	• Custom relational table with unique Auto-Numbering for tickets. • Stores all submitted variables, payment details, and address. • Maintains real-time Request State (Submitted, Waiting Approval, Approved, Rejected). • Integrates with sysapproval_approver for approval audit tracking.

Communications Tier	ServiceNow Outbound Email Engine	• Integrates with platform SMTP service to dispatch automated notifications. • Generates dynamic milestone emails containing ticket Auto-Number, user details, and approval decisions at Steps 3, 7, and 10.
Security & Governance Tier	Platform Access Control Lists (ACLs)	• Role-Based Access Control (RBAC) enforced via platform ACLs. • Requesters have read access to their own submissions. • Approvers and administrators have governed update/approval permissions. • Full execution logs available in Flow Execution Details for audit compliance.

3. Solution Design Principles & Quality Attributes

- **Zero-Code Maintainability:** The entire workflow is built using native Flow Designer actions without writing custom Script Includes or Business Rules, reducing technical debt and enabling visual debugging.
- **Data Integrity & Validation:** Catalog UI Policies prevent blank submissions by validating mandatory fields (Connection Type, Phone, Email, Payment Mode) before records are created in Database Tables.
- **End-to-End Traceability:** Every request generates a unique Auto-Numbered record linked with approval history in sysapproval_approver and logged email dispatches for complete audit readiness.
- **Extensible Foundation:** The architecture easily accommodates future enterprise enhancements, such as connecting to network configuration tools (Cisco DNA Center, Ansible) via Integration Hub spokes.